Character Array Practice Questions

Write a function for each of the following questions. The function takes one or more arrays of characters and **returns** the answer. The function must return the result, not print it.

> **Quick reminder:** `return result;` is correct. `print(result)` inside the function is not.

1 Count the vowels and consonants. Return both as [vowels, consonants].

Input to the function: `['a','p','p','l','e']`
Return from the function: `[2, 3]`

2 Return how many times a given character appears.

Input to the function: `['b','a','n','a','n','a'] and 'a'`
Return from the function: `3`

3 Return the array with every letter in capitals.

Input to the function: `['c','o','d','e']`
Return from the function: `['C','O','D','E']`

4 Return true if the array contains at least one digit.

Input to the function: `['p','a','s','s','1']`
Return from the function: `true`

5 Return the first and the last character as a new array.

Input to the function: `['m','a','n','g','o']`
Return from the function: `['m','o']`

6 Return how many capital letters are in the array.

Input to the function: `['I','n','d','i','a','N','o','1']`
Return from the function: `2`

7 Password check: return true if the password has at least 8 characters and at least one digit.

Input to the function: `['s','e','c','r','e','t','9','x']`
Return from the function: `true`

8 Return the array with every space replaced by an underscore '_'.

Input to the function: `['h','i',' ','t','h','e','r','e']`
Return from the function: `['h','i','_','t','h','e','r','e']`

9

Return the number of words. Words are separated by single spaces.

Input to the function: `['I',' ','l','o','v','e',' ','J','a','v','a']`
Return from the function: `3`

10

Given a full name, return the initials in capitals.

Input to the function: `['r','a','h','u','l',' ','d','r','a','v','i','d']`
Return from the function: `['R','D']`

11

Return the array with all vowels removed.

Input to the function: `['p','r','o','g','r','a','m']`
Return from the function: `['p','r','g','r','m']`

12

Return the letter that comes last in alphabetical order.

Input to the function: `['d','o','g','s']`
Return from the function: `'s'`

13

Given the grades of a class, return how many students got 'A'.

Input to the function: `['A','B','A','C','A','B']`
Return from the function: `3`

14

Return the sum of all the digit characters, ignoring letters.

Input to the function: `['a','3','b','7','2']`
Return from the function: `12`

15

Given the traffic signal colours seen in a day ('R', 'G', 'Y'), return the counts as [red, green, yellow].

Input to the function: `['R','G','G','Y','R','G']`
Return from the function: `[2, 3, 1]`

16

Mobile number check: return true if there are exactly 10 characters and all are digits.

Input to the function: `['9','8','4','5','0','1','2','3','4','5']`
Return from the function: `true`

17

Return the array with each letter moved one step forward in the alphabet ('z' becomes 'a').

Input to the function: `['c','a','t','z']`
Return from the function: `['d','b','u','a']`

18

Return the positions (indexes) where a given character appears.

Input to the function: `['h','e','l','l','o'] and 'l'`
Return from the function: `[2, 3]`

19

Return true if the word reads the same forwards and backwards.

Input to the function: `['m','a','d','a','m']`
Return from the function: `true`

20

Given two words of the same length, return at how many positions they have the same character.

Input to the function: `['c','a','r','d'] and ['c','o','r','e']`
Return from the function: `2`

21

Given a student's quiz answers and the answer key, return the score (1 mark per correct answer).

Input to the function: `['A','C','B','D','A'] and ['A','B','B','D','C']`
Return from the function: `3`

22

Return the characters in the first array that are not present in the second array.

Input to the function: `['a','b','c','d'] and ['b','d']`
Return from the function: `['a','c']`

23

Given a first name and a last name, return the full name with a space in between.

Input to the function: `['r','a','m'] and ['k','u','m','a','r']`
Return from the function: `['r','a','m',' ','k','u','m','a','r']`

24

Return true if both arrays have exactly the same letters, in any order.

Input to the function: `['s','i','l','e','n','t'] and ['l','i','s','t','e','n']`
Return from the function: `true`

25

Votes from three polling booths are given ('A' or 'B' for each voter). Return the candidate with the most votes overall.

Input to the function: `['A','B','A'], ['B','B','A','B'] and ['A','A']`
Return from the function: `'A'`